

Phishing 2026: AI, Asymmetry and the Modern Defence Architecture

Module Focus : Phishing Awareness and Defence Protocol

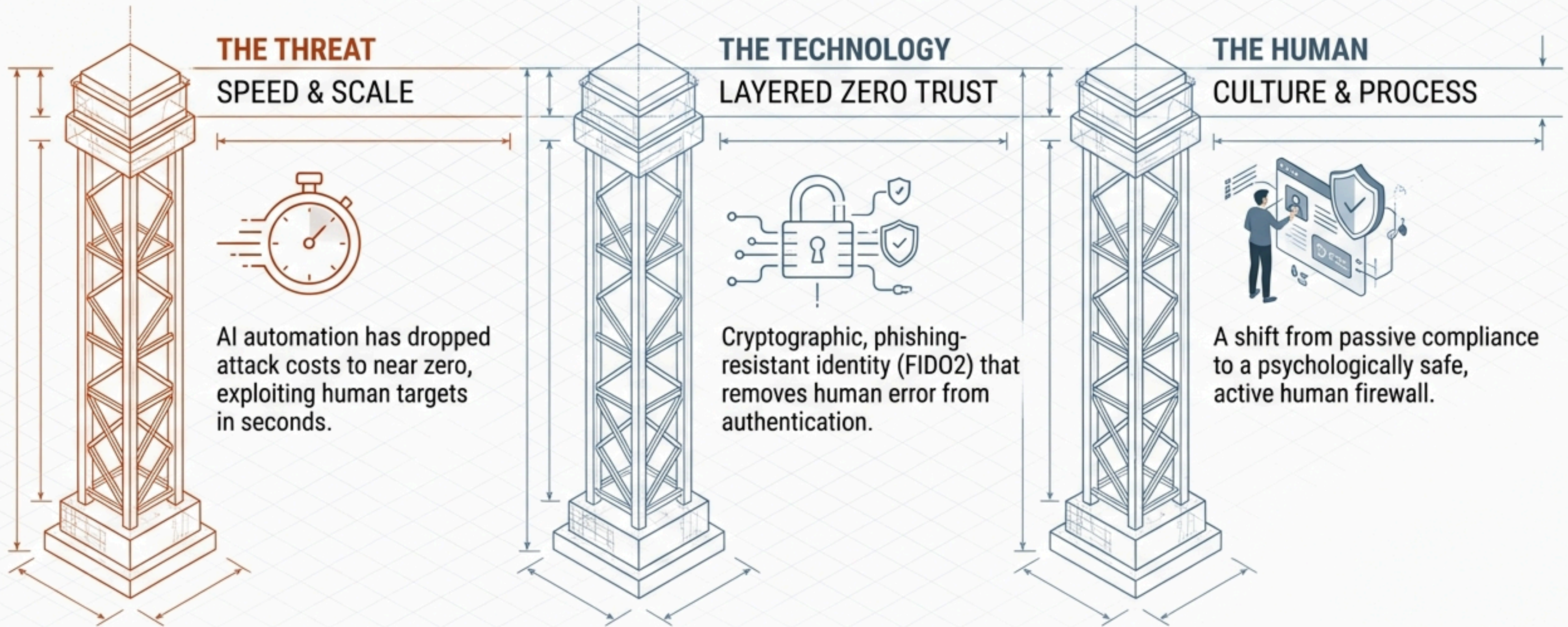
Objective : Transition from passive Consumers to active threat detectors

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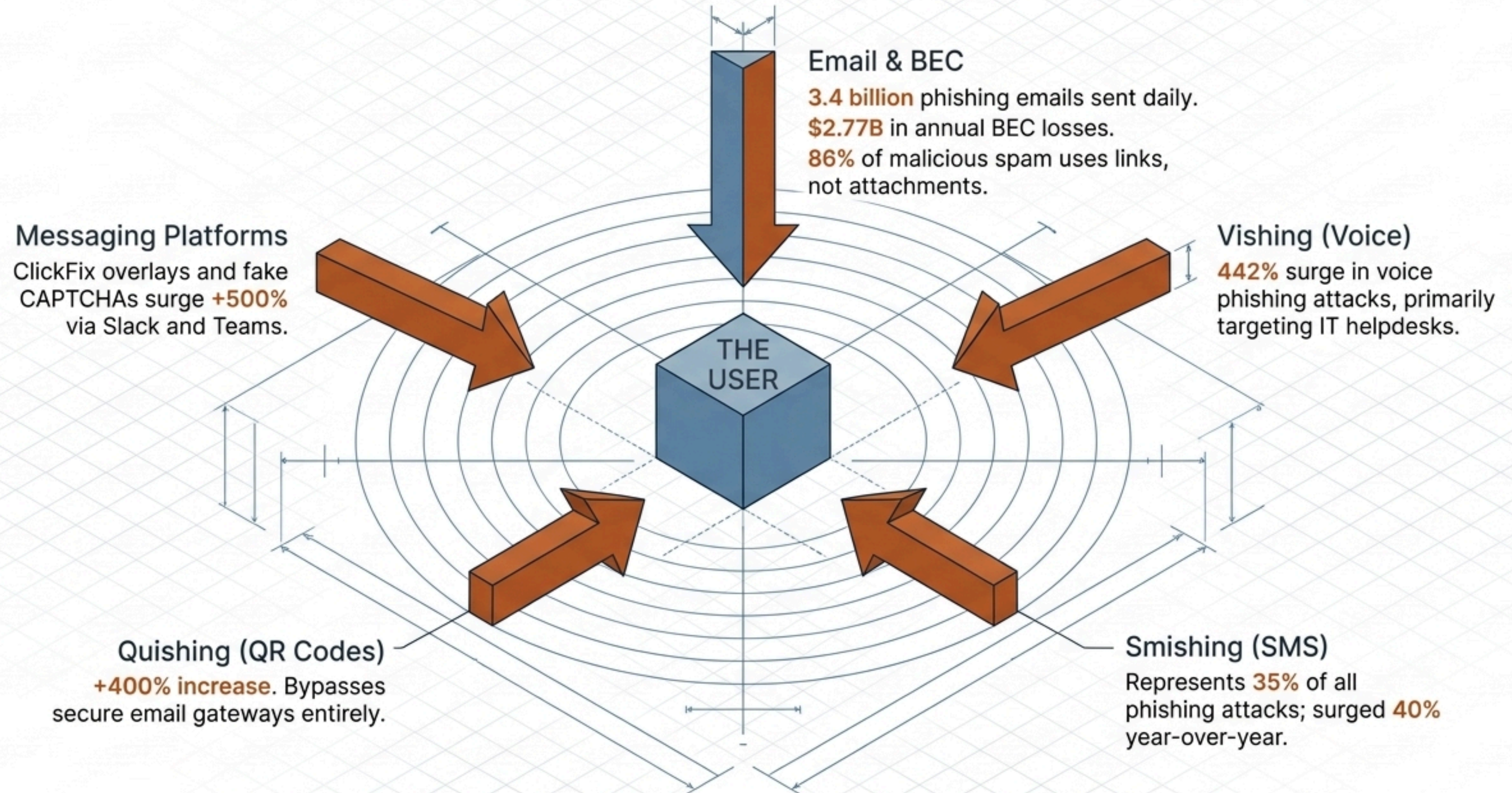


THE ASYMMETRIC WARFARE PARADIGM

Phishing is no longer a mass-email annoyance; it is an AI-automated, multi-channel breach vector that bypasses traditional MFA. Defence requires a paradigm shift from passive filtering to proactive, identity-centric security.



THE FRAGMENTED ATTACK SURFACE



THE AI ACCELERANT AND DEEPFAKE ECONOMICS

Dimension	Traditional Phishing	AI-Driven Phishing
1 Volume & Quality	High volume, poor grammar.	86% of all phishing is AI-generated, featuring flawless prose.
2 Attack Cost	Requires manual human reconnaissance.	95%+ cost reduction for attackers.
3 Efficacy	~2.7% baseline click rate.	54% click rate on AI spear-phishing.
4 Format	Text-only emails.	Cloned Voice & Video. Only 0.1% of people can identify a deepfake.

The \$25M Deepfake

Arup CFO Scam: A finance employee authorised \$25.6M in wire transfers after a video call where every other participant—including the “CFO”—was an AI deepfake.

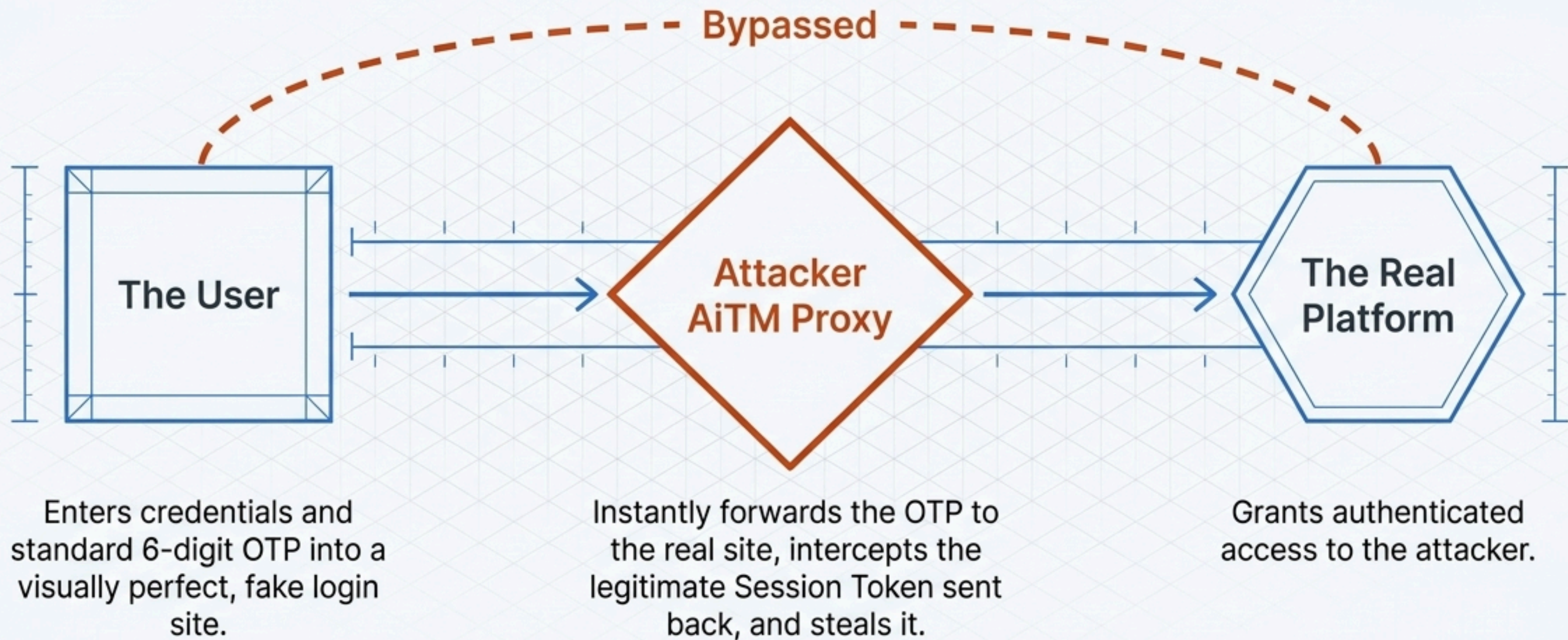
The Speed Asymmetry: The 21-Second Problem

If our defence relies solely on human reporting, the attacker has a 27-minute uncontested head start.



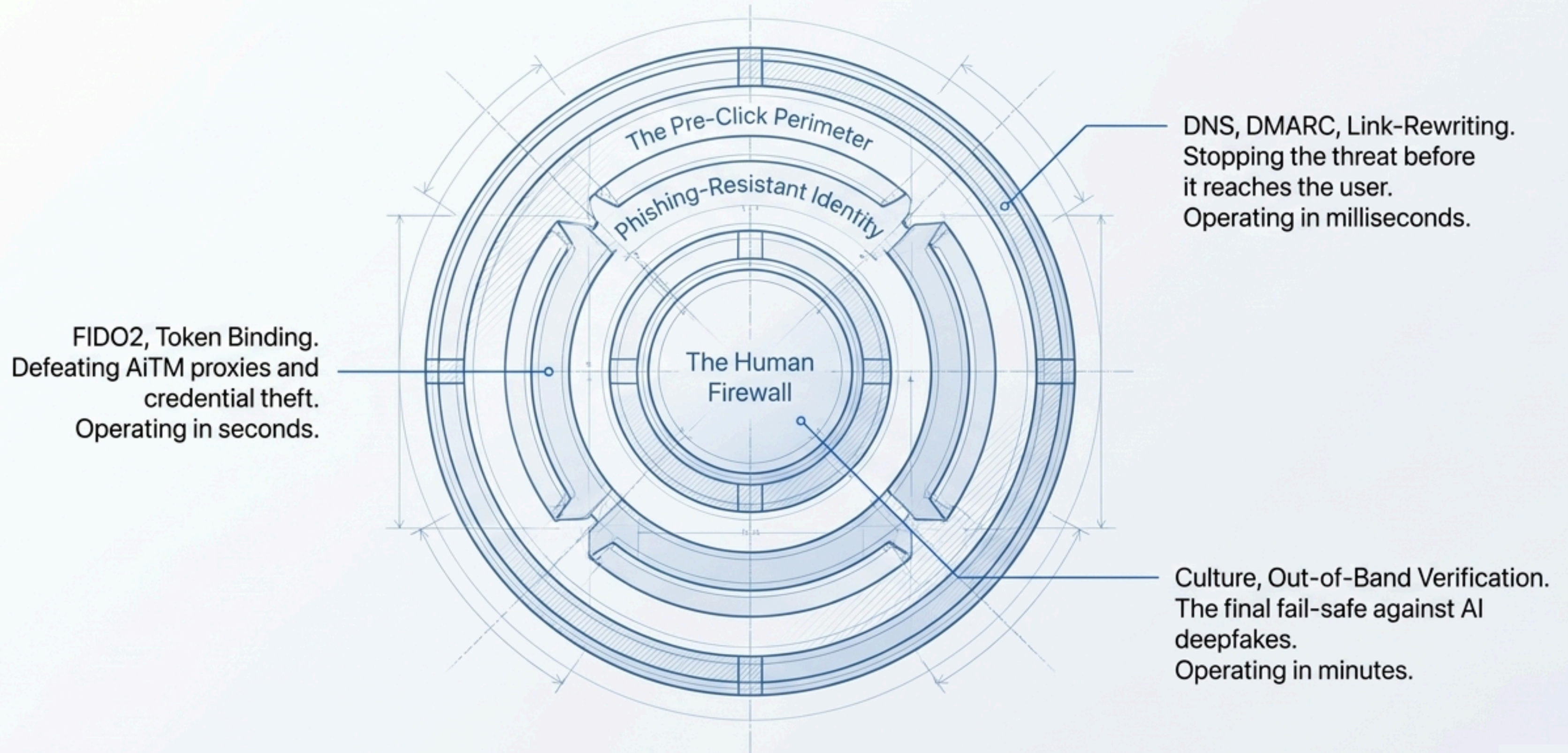
The MFA Illusion: Adversary-in-the-Middle (AiTM)

80% of MFA-bypass breaches now use session-token theft. Commodified AiTM kits (like Tycoon 2FA, Evilginx) cost just \$120–\$350 a month.



The Defence-in-Depth Architecture

If the attacker operates in seconds, our technical defence must operate in milliseconds, buying time for our human defence to verify.



Layer 1: The Pre-Click Perimeter

39,000 phishing emails
sent globally per second.



DMARC (p=reject)

Blocks domain spoofing and prevents attackers from impersonating our exact brand.



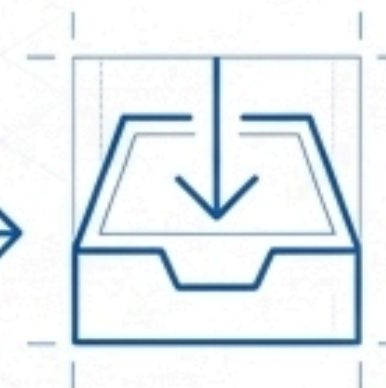
URL Sandboxing

Detonates suspicious links in a secure environment. Vital because 86% of malicious spam relies on URLs.



Protective DNS

Blocks network-level traffic to malicious domains, neutralizing attacks even from SMS or personal channels.



Layer 2: Phishing-Resistant Identity

To defeat session-token theft, we must mandate FIDO2 passkeys and hardware keys.

Legacy MFA



Standard MFA relies on the human verifying the site.

User sees visually perfect fake URL -> Inputs OTP code
-> Breach.

FIDO2 / Passkeys



FIDO2 relies on cryptography verifying the site.

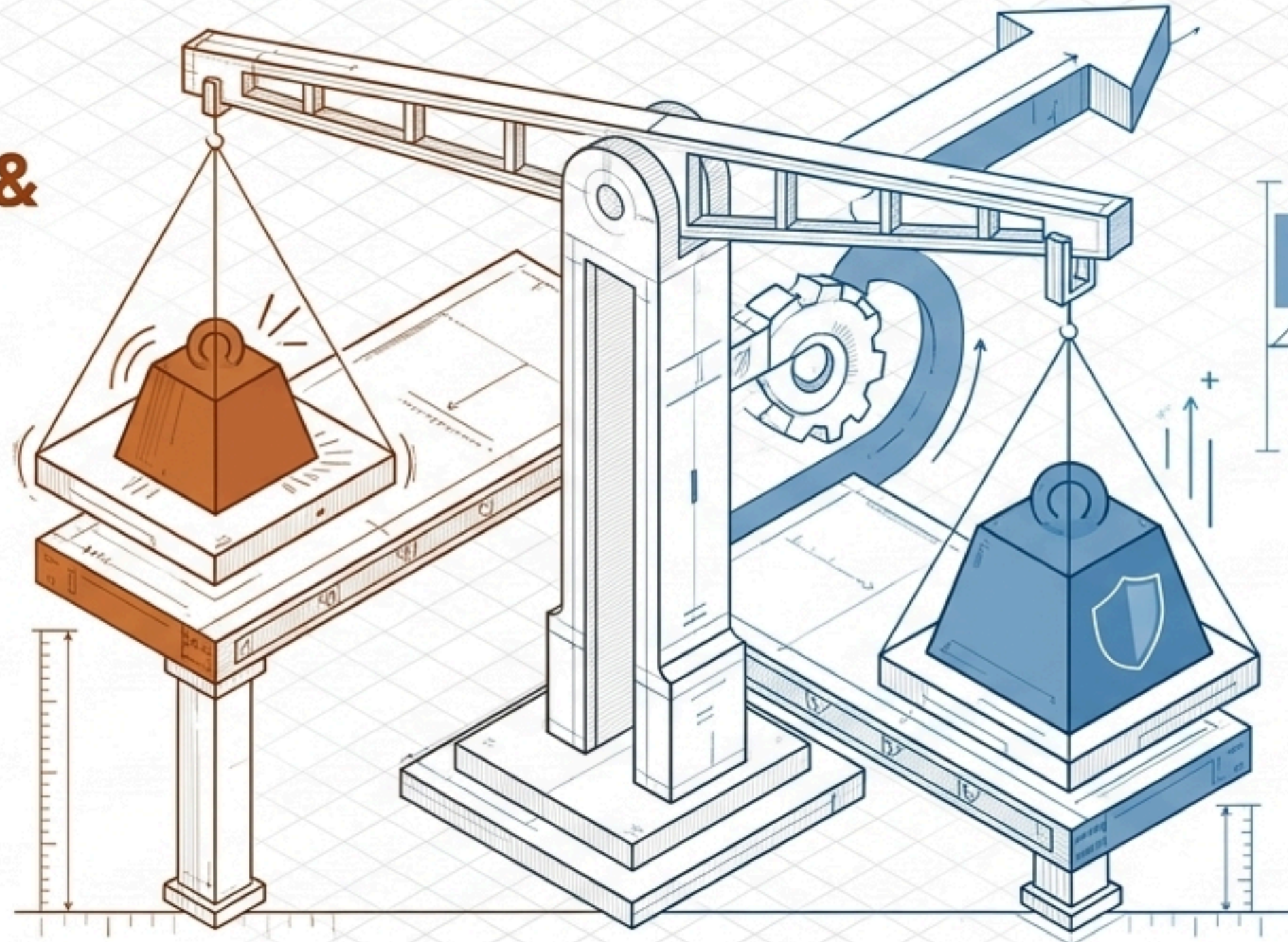
Device cryptographically detects hidden domain
mismatch -> Silently refuses to authenticate -> Blocked.

Layer 3: The Human Firewall & Culture Shift

60% of all breaches involve the human element. We cannot punish users for falling for perfect AI; we must reward them for reporting anomalies instantaneously.

Culture of Compliance & Punishment

- Drives fear and anxiety.
- Results in hiding accidental clicks.
- Creates dangerous, delayed reporting.

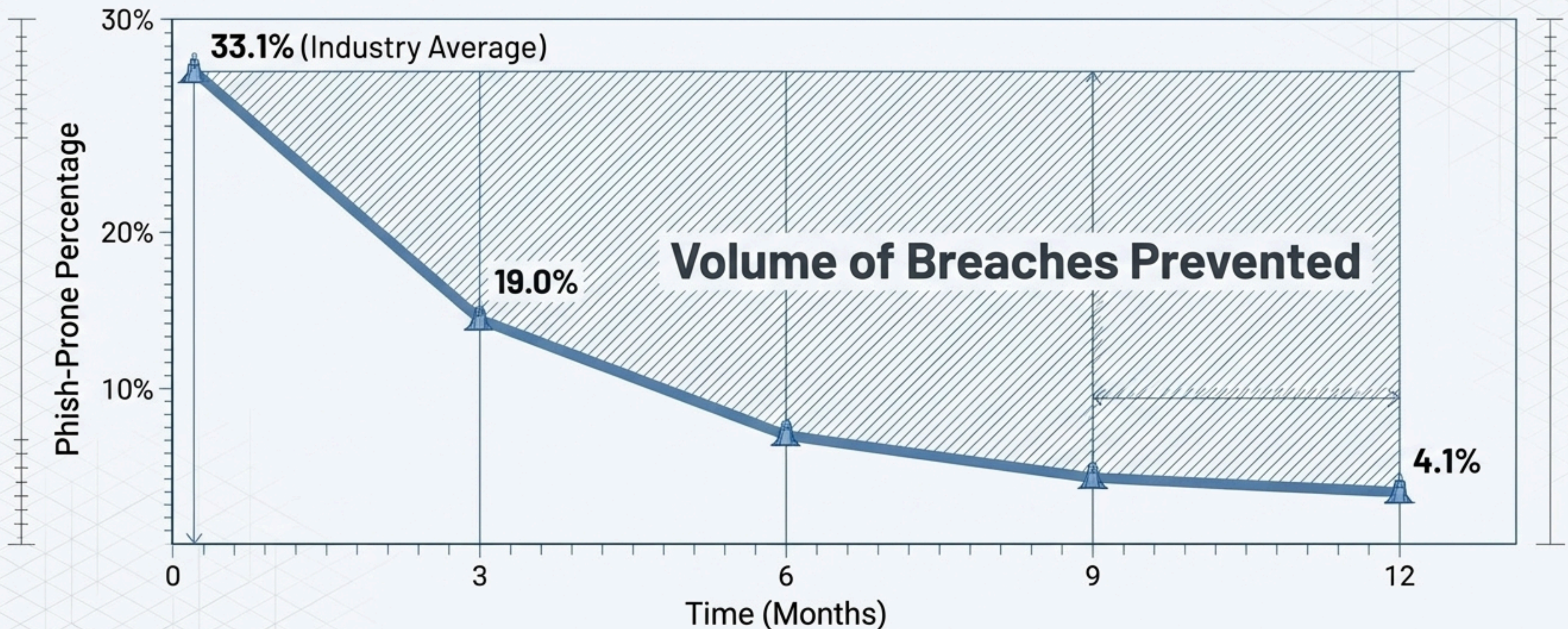


Culture of Psychological Safety

- Drives proactive vigilance.
- Emphasises immediate, non-punitive reporting.
- Fosters gamified threat hunting.

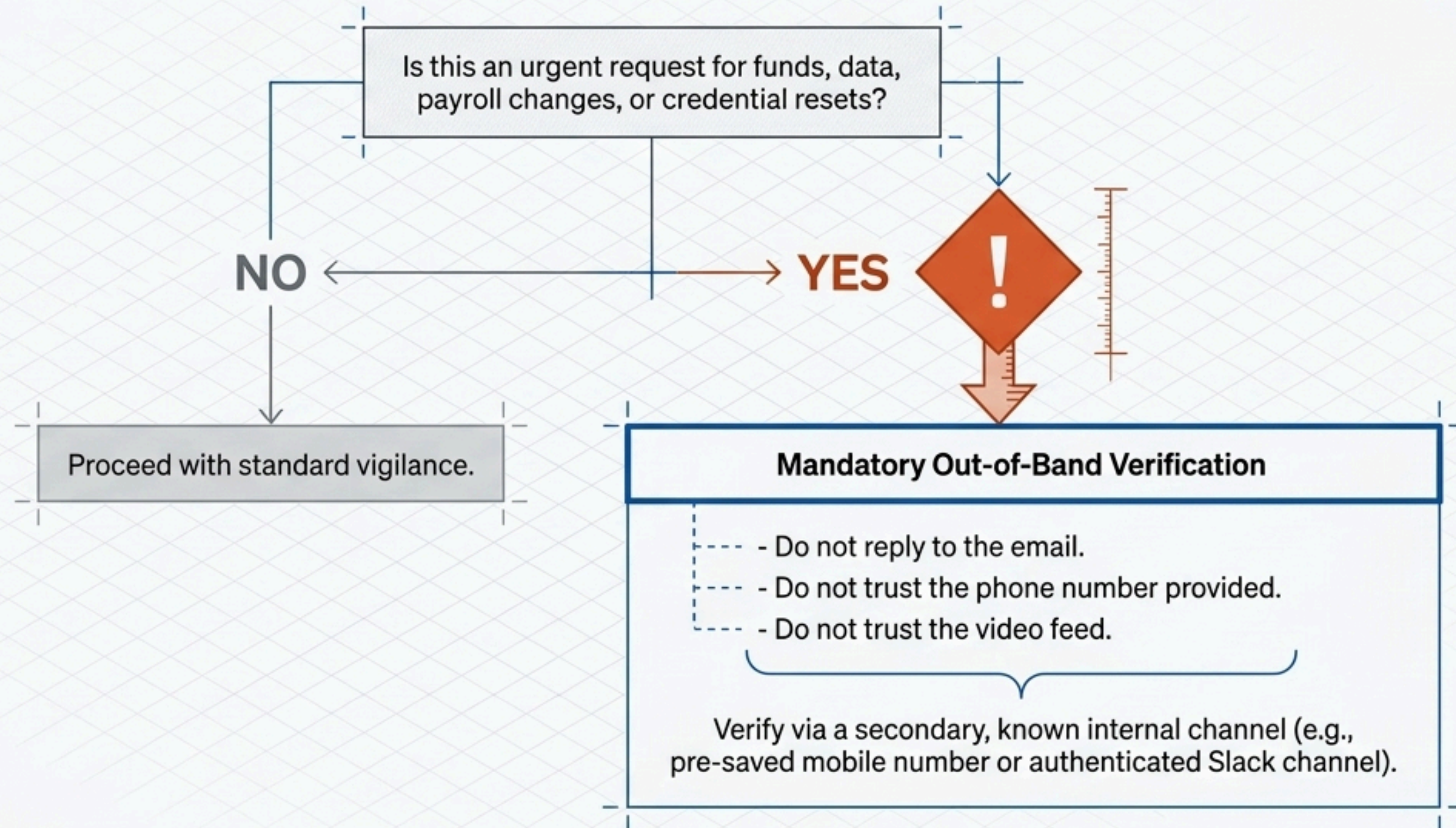
The ROI of Continuous Simulation

Continuous phishing simulation transforms employees from vulnerabilities into active sensors, resulting in an **86%** reduction in susceptibility.



Process Over Vigilance: Out-of-Band Verification

You cannot train a human to spot a perfect AI deepfake. Therefore, defence against scenarios like the \$25M Arup scam relies entirely on structural process.

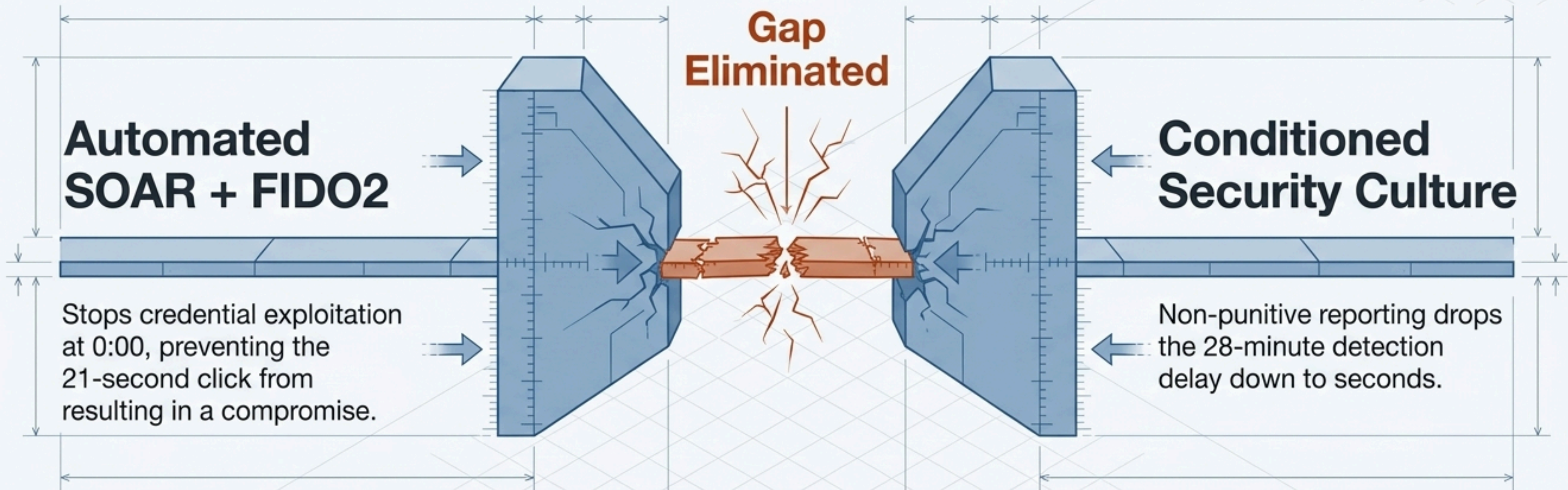


The 2026 Defence Matrix

	Pre-Click Filtering (DMARC/DNS)	Phishing-Resistant Identity (FIDO2)	Out-of-Band Process	Continuous Simulation
BEC / Wire Fraud	✓ Mitigated (Domain Protection)	✓	✓	✓
AiTM Proxy	✓	✓ Neutralised (Cryptographic Block)	✓	✓
AI Deepfakes	✓	✓	✓ Neutralised (Channel Break)	✓
Smishing / QR Codes	✓	✓	✓	✓ Mitigated (Conditioned Skepticism)

Closing the Attacker's Window

The asymmetric advantage returns to the defender.



Strategic Mandates for Leadership

Mandate Phishing-Resistant Identity

Audit all MFA configurations and mandate FIDO2/Passkey adoption, prioritising executives, finance, and IT administrators.

Enforce Out-of-Band Verification

Implement strict, unbypassable out-of-band verification policies for all financial transfers, data exports, and helpdesk credential resets.

Fund Continuous Simulation

Commit resources to high-frequency, multi-channel simulation programs paired with a strictly non-punitive reporting culture.

Defeating AI requires systematic resilience, not human perfection.

INTERACTIVE ASSESSMENT & PHISHING QUIZ

ASSESSMENT OVERVIEW

- **Target Objective:** Put your threat recognition skills to the test with real-world phishing scenarios.
- **Evaluation Format:** 8 interactive scenario-based questions featuring email, URL, and messaging traps.
- **Benchmark Score:** Achieve 6 or more correct answers (75%+) to earn Phishing-Aware Certification.
- **Immediate Feedback:** Review detailed explanations after each question to reinforce key defensive concepts.



THE DEFENCE PROTOCOL: CONCLUSION & ACTION PLAN

STOP

Do not click or reply right away.

THINK

Is it urgent, unexpected or too good to be true?

VERIFY

Use an official site or phone number.

REPORT

Tell IT, the platform or your bank.

STRATEGIC PILLAR 1: TECHNICAL & CULTURAL HYGIENE

- Phishing-Resistant MFA: Upgrade to FIDO2 / Passkeys to defeat AiTM proxying.
- Domain Authentication: Enforce strict DMARC (p=reject) & DNS filtering.
- Continuous Simulation: Lower susceptibility from 33% to <4% via ongoing training.

STRATEGIC PILLAR 2: EMERGENCY COMPROMISE ACTION

- Disconnect & Contain: Turn off Wi-Fi/Ethernet immediately if a file was executed.
- Credential Reset: Change master passwords and revoke active session tokens.
- Non-Punitive Reporting: Escalate instantly to IT/SOC - speed stops lateral movement.



THANK YOU!

